1) Determinazione specifiche di progetto

Incognite: G_F; K_c da ε_r; K_c da d_a e/o d_p; ω_c da d_p e/o d_s; ζ, $T_{p,0}$, $S_{p,0}$ da $\hat{s}$; ω_c da t_r; ω_c da $t_{s,\varepsilon\%}$; ν da d_a e/o d_p; M_S^{LF}, M_T^{HF}.

a) Calcolo G_F

$$G_F = \frac{1}{G_s\,K_d} \quad \text{se } \nu + p > 0$$

b) Calcolo ν e K_c da ε_r

se $\varepsilon_r = 0 \to \nu > h - p, |K_c| = \forall$
se $\varepsilon_r \neq 0 \to \nu \geq h - p$

$$|K_c| \geq \begin{cases} \left|\dfrac{R_0\,K_d^2}{\varepsilon_r\,K_p\,G_a}\right| & \text{se } \nu + p \neq 0 \\ \dfrac{\dfrac{R_0\,K_d^2}{\varepsilon_r} - K_d}{K_p\,G_a} & \text{se } \nu + p = 0 \end{cases}$$

K_p = guadagno di G_p in cost. tempo. (p: poli nell'origine di G_p espresso in cost. tempo)

$\texttt{zpk\_gp = zpk(Gp);}$

$\texttt{zpk(zpk\_gp.Z, zpk\_gp.P, zpk\_gp.K,}$ 'DisplayFormat', 'timeconstant')

c) Calcolo K_c da d_a

$$|e_\infty^{d_a}| = \left|\lim_{t\to\infty} e_{d_a}(t)\right| = \left|\lim_{t\to\infty} y_{d_a}(t)\right| = \left|\lim_{s\to 0} s\,G_{d_a y}(s)\,d_a(s)\right|$$

$$|K_c| \geq \left|\frac{D_{a,0}}{\varepsilon_{d_a}\,G_a\,G_s\,G_F}\right| \quad \text{se } \varepsilon_{d_a} \neq 0 \;\Rightarrow\; \nu \geq h_a$$

$$|K_c| = \forall \quad \text{se } \varepsilon_{d_a} = 0 \;\Rightarrow\; \nu > h_a$$

d) Calcolo K_c da d_p (polinomiale) o ω_c/M_S^{LF} (sinusoidale)

Polinomiale:

$$|e_\infty^{d_p}| = \left|\lim_{t\to\infty} e_{d_p}(t)\right| = \left|\lim_{t\to\infty} y_{d_p}(t)\right| = \left|\lim_{s\to 0} s\,G_{d_p y}(s)\,d_p(s)\right|$$

$$|K_c| \geq \left|\frac{D_{p,0}}{\varepsilon_{d_p}\,G_a\,G_s\,G_F\,K_p}\right| \quad \text{se } \varepsilon_{d_p} \neq 0 \;\Rightarrow\; \nu \geq h_p - p$$

$$|K_c| = \forall \quad \text{se } \varepsilon_{d_p} = 0 \;\Rightarrow\; \nu > h_p - p$$

Sinusoidale:

$$|e_\infty^{d_p}| = \left|G_{d_p y}(j\omega_p)a_p \sin(\omega t + \varphi_p)\right| \leq |S(j\omega_p)|$$

$$M_S^{LF} = \frac{\rho_p}{a_p}, \qquad M_{S,\,dB}^{LF} = 20\log_{10} M_S^{LF}$$

$$\omega_L = \omega_p \cdot 10^{-M_{S,\,dB}^{LF}/40}, \qquad \omega_c \geq 2\,\omega_L$$

e) Calcolo ω_c/M_T^{HF} da d_s (sinusoidale)

$$|e_\infty^{d_s}| = \left|G_{d_s y}(j\omega_s)a_s \sin(\omega t + \varphi_s)\right| \leq |T(j\omega_s)|$$

$$M_T^{HF} = \frac{\rho_s\,G_s}{a_s}, \qquad M_{T,\,dB}^{HF} = 20\log_{10} M_T^{HF}$$

$$\omega_H = \omega_s \cdot 10^{M_{T,\,dB}^{HF}/40}, \qquad \omega_c \leq \frac{\omega_H}{2}$$

f) Calcolo ζ, $T_{p,0}$, $S_{p,0}$ da $\hat{s}$

$$\zeta \geq \frac{|\ln(\hat{s})|}{\sqrt{\pi^2 + \ln^2(\hat{s})}}$$

$$T_{p,0} \leq \frac{1}{2\zeta\sqrt{1-\zeta^2}}$$

$$S_{p,0} \leq \frac{2\zeta\sqrt{2 + 4\zeta^2 + 2\sqrt{1+8\zeta^2}}}{\sqrt{1+8\zeta^2} + 4\zeta^2 - 1}$$

g) Calcolo ω_c da t_r e $\varepsilon_{s,\%}$

$$t_r: \quad \omega_c \geq \left(\frac{\pi - \arccos(\zeta)}{t_r\sqrt{1-\zeta^2}}\right)\sqrt{\sqrt{1+4\zeta^4} - 2\zeta^2}$$

$$t_{s,\alpha\%}: \; \omega_c \geq \left(\frac{|\ln(\alpha\%)|}{\zeta\,t_{s,\alpha\%}}\right)\sqrt{\sqrt{1+4\zeta^4} - 2\zeta^2}$$

h) Conclusioni

Scelgo ν più grande tra i ν che ho trovato e prendo il K_c più grande tra quelli che ho trovato con quel ν.

2) Progetto del Sistema di Controllo

Passo 1: Studio del segno di K_c e stabilizzabilità

a. Scelgo un segno e calcolo $L_{in}(s)$:

$$L_{in}(s) = \frac{K_c}{s^\nu} \cdot G_F\,G_a\,G_p\,G_s$$

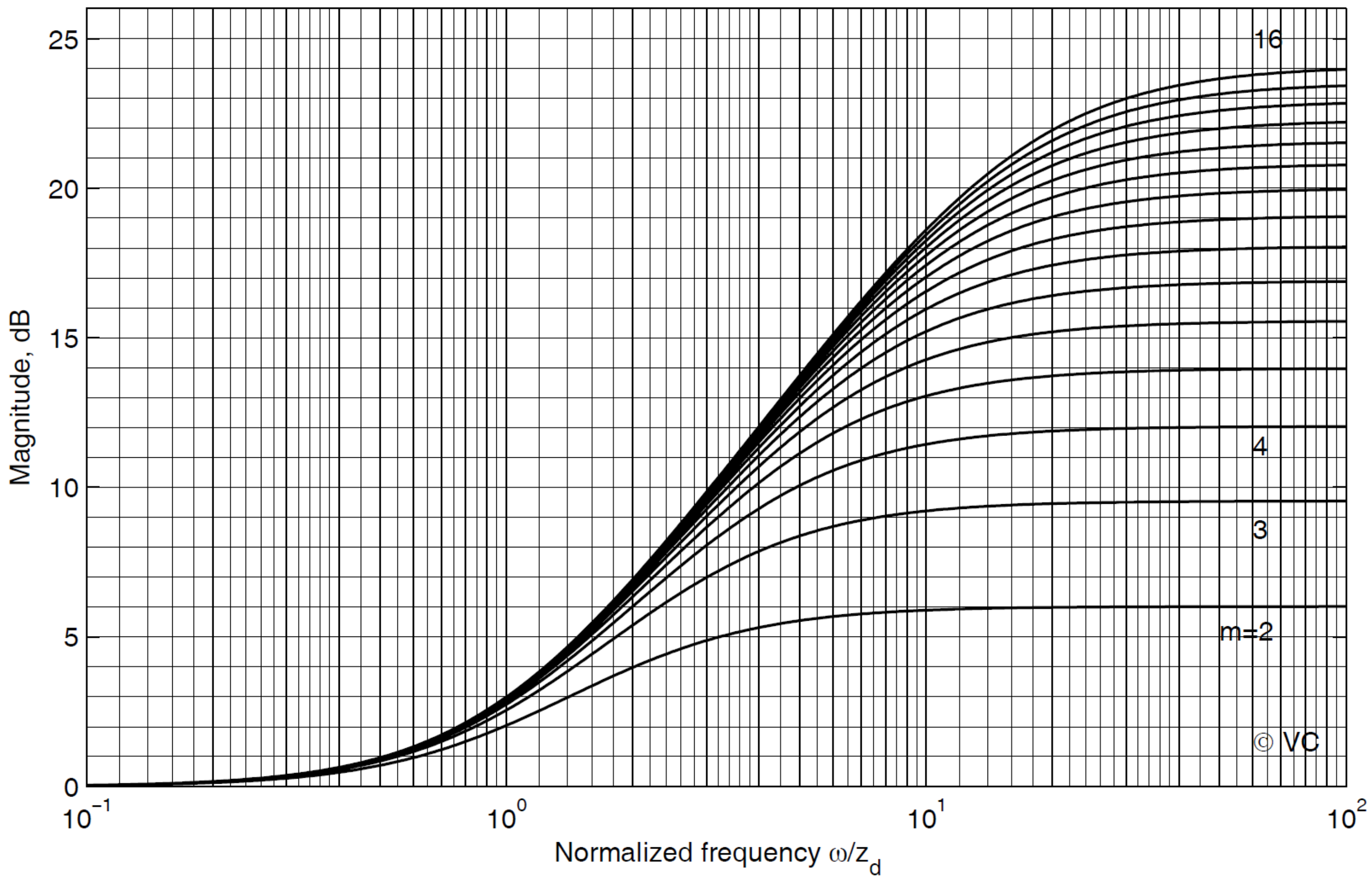
b. Tracciare il diagramma di Nyquist di $L_{in}(s)$ e calcolo $P_{cl} = N + P_{ol}$, dove:
N: intersezioni semiretta
P_{ol}: poli con $Re(p_i) > 0$ (di solito nullo)

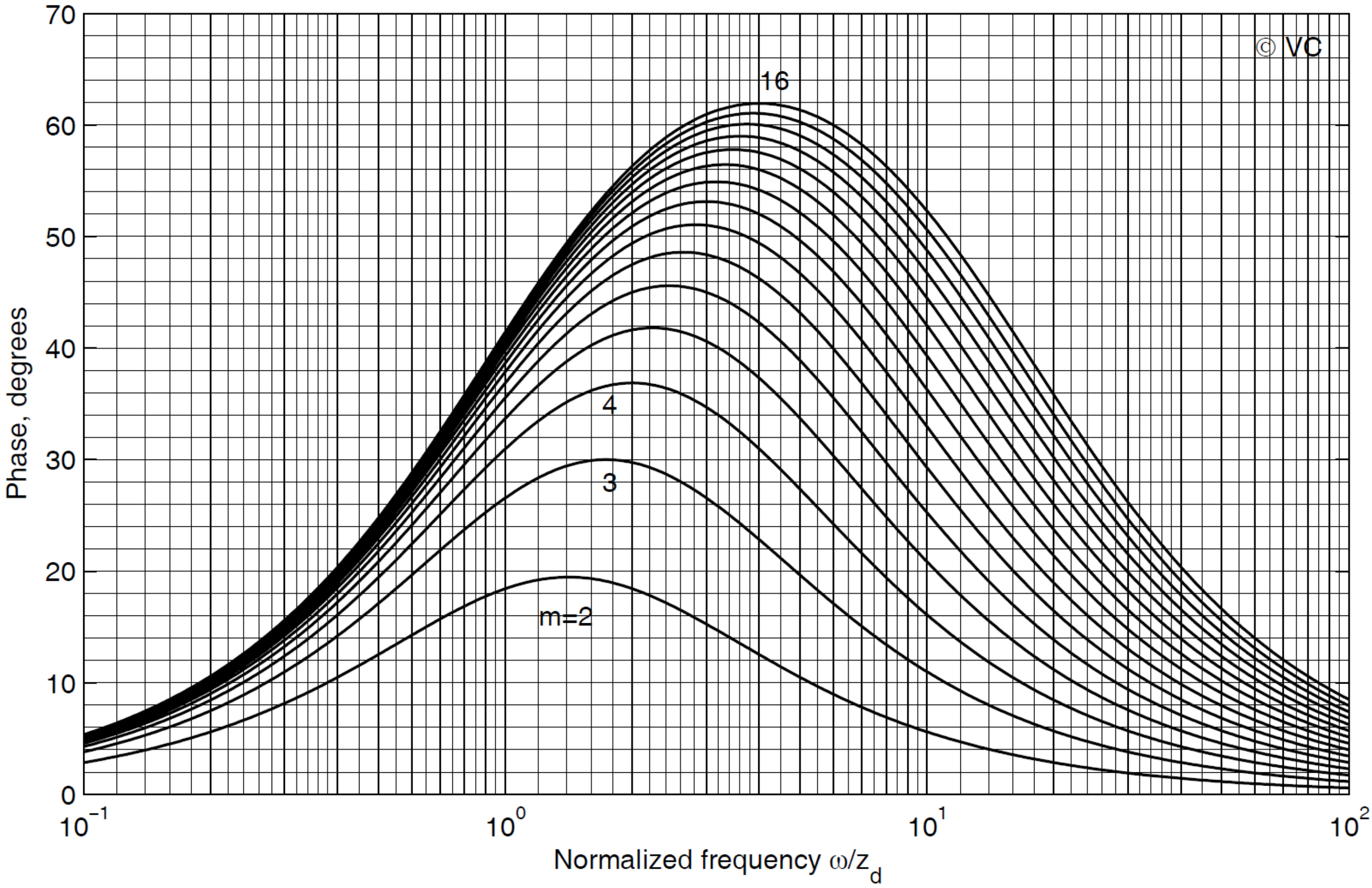
- P_{cl} pari → **Stabilizzabile**
- P_{cl} dispari → **Cambio segno di K_c**

c. Scelgo K_c poco più grande del vincolo (se c'è).

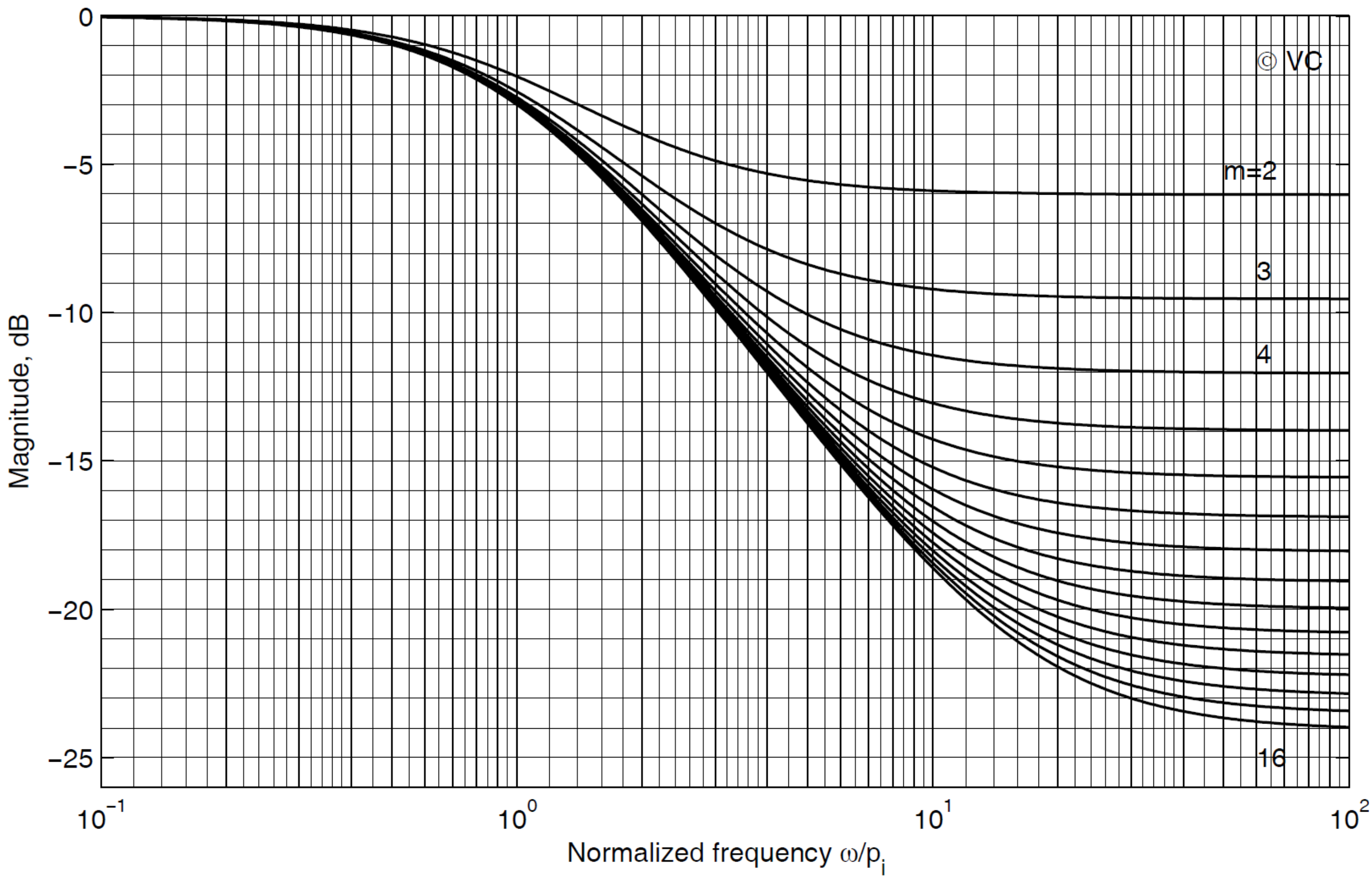
$$[N_{L_{in}}, D_{L_{in}}] = \texttt{tfdata}(L_{in}, \text{“v”})$$
$$\texttt{myngridst}(T_{p,0}, S_{p,0})$$
$$\texttt{nyquist1}(N_{L_{in}}, D_{L_{in}})$$

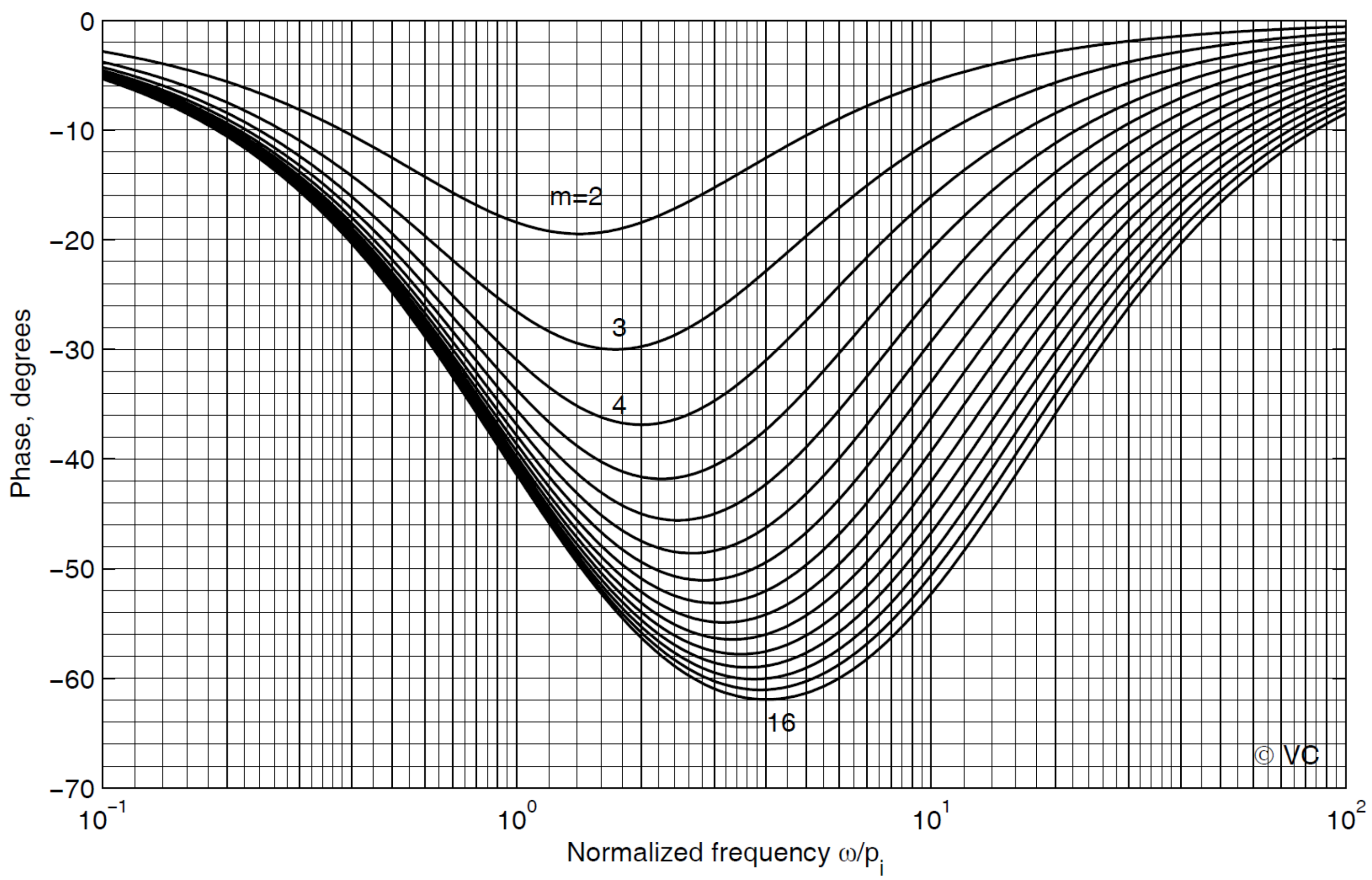
Phase LEAD network





Phase LAG network





Zero network